

Deep Video Compression with Scaled Hierarchical Bi-directional Motion Model

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Abstract

Deep video compression has attracted increasing attention in recent years due to its end-to-end optimization ability. However, most existing neural video compression (NVC) models focus on incorporating sophisticated motion or residual coding networks for successive frames leveraging spatial-temporal redundancy removal, neglecting the efficient motion representation and essential structure of scaled prediction for motion dynamics. To resolve this problem, this paper proposed a novel model, named scaled hierarchical bi-directional prediction structure, which effectively captures temporal correlation among frames considering the quality variation when managing the reference frames. This paper first introduces parameter-shared motion codecs and efficient information fusion strategies to obtain predictive features more precisely. Subsequently, scaled motions from temporal contexts are learned as bi-directional prior for motion representation. Additionally, the concept of *trustworthy motion modeling* is proposed to represent the effectiveness of reference information, measuring the reliability of predictive accuracy in complex motions, camera rotations and occlusions. Extensive experimental results demonstrate that our approach offers significant advantages over state-of-the-art bi-directional NVC models in coding efficiency. The proposed method has been adopted as the latest reference model by Moving Picture, Audio, and Data Coding by Artificial Intelligence (MPAI) end-to-end video coding (EEV) standard. The code is available at: https://github.com/yefeng00/DVC_with_Scaled_Hierarchical_Bi_directional_Motion_Model.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Machine learning*.

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This work was supported in part by the National Key R&D Program of China in part by 2023YFB2809603, in part by the NSFC 62371008, and 62101007, and in part by the Fundamental Research Funds for the Central Universities.

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MM '24, October 28-November 1, 2024, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-0686-8/24/10

<https://doi.org/10.1145/3664647.3685524>

Keywords

End-to-end learned video compression, standardization, MPAI, scaled hierarchical motion model, bi-directional prediction

ACM Reference Format:

Feng Ye, Li Zhang, and Chuanmin Jia. 2024. Deep Video Compression with Scaled Hierarchical Bi-directional Motion Model. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24)*, October 28-November 1, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3664647.3685524>

1 Introduction

Since the hybrid coding framework combining prediction and transform was proposed, the framework has achieved great success in the research field of image-video coding and is still in use today. However, traditional coding standards optimize each module independently, which limits the improvement of the overall compression performance. With the development of neural networks, their excellent nonlinear modeling capability and overall optimization feasibility have become emerging research topics in the field of video coding [10, 12]. The neural network video compression (NVC) models have become a brand new development direction due to their good compression performance, flexible design method and convenient hardware deployment. Most existing methods [3, 11] are based on hybrid coding architectures for low-latency P-frame coding, including uni-directional temporal prediction and deep residual coding. It is shown in [7] that residual coding is suboptimal from the perspective of Shannon entropy.

Compared to the rapid development of learned P-frame coding, learned B-frame coding, which allows bidirectional referencing for higher coding efficiency, is an understudied topic. Representative bi-directional coding schemes [1, 4, 14, 16] utilize frame interpolation and explicit motion modeling with residual or conditional coding. In contrast, the model proposed in [16] treated video coding as image interpolation, employing deep image generation and interpolation for hierarchical coding of B-frames. Alexandre et. al. used a low-resolution base layer containing interpolated frames instead of a motion encoder to improve compression performance [1].

However, most of the existing neural video compression (NVC) models focus on incorporating complex motion or residual coding networks for consecutive frames by utilizing spatial-temporal redundancy removal techniques, while ignoring the basic structure of efficient motion representation and hierarchical prediction of motion dynamics. The proposed approach introduces a scaled hierarchical bi-directional prediction structure, which effectively captures temporal correlation among frames considering the quality variation when managing the reference frames. In summary, our contributions are:

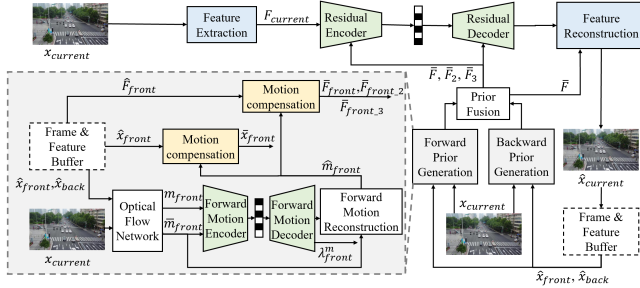


Figure 1: The framework of the proposed NVC with scaled hierarchical bi-directional motion model. λ_{front}^m refers to the trustworthy motion confidence described in Section 2.4.

- We propose a novel bi-directional NVC model using scaled hierarchical motion, which has parameter-shared motion codecs and scaled bi-directional motion representation prior to temporal contexts.
- We propose the concept of trustworthy motion modeling to represent the effectiveness of reference information, improving the reliability of predictive accuracy in complex motions, camera rotations, and occlusions.
- The robust learning strategy and rate-distortion optimization techniques are also proposed. Extensive experimental results demonstrate that our approach offers significant advantages over state-of-the-art NVC models in coding efficiency.

2 Design Overview

2.1 Overview

To achieve higher coding efficiency, our codec employs a scaled hierarchical bi-directional prediction structure. The framework of our model is illustrated in Fig. 1. It is noted that our model focuses on the efficient motion representation and essential structure of scaled prediction for motion dynamics to achieve a better understanding of the temporal properties of videos.

For coding each frame $x_{current}$, our coding scheme contains three steps: prior information generation, predictive feature integration and residual coding. At first, optical flow network is used to estimate both forward and backward motion vectors m_{front} and m_{back} separately. Then motion encoding, decoding and compensation are also accomplished individually to get bidirectional predictive features. After that, efficient information fusion strategies are utilized to obtain final precise prior features. At last, reconstructed frame $\hat{x}_{current}$ is generated via a residual encoder, decoder and frame reconstructor.

2.2 Motion Coding and Information Fusion

Instead of encoding the bidirectional motion vectors with only one codec, we propose to encode both sides of the motion separately, using two parameter-shared codecs. With this design, the efficient motion coding module of the unidirectional coding scheme can be directly migrated to this framework. The extra motion codec facilitates the acquisition of more accurate motion prediction information, leading to a better understanding of the temporal domain characteristics of the videos.

How to effectively fuse the motion information of both sides will significantly affect the efficiency of residual coding. To this end, we extract three layers of features for both reference frames and reconstructed motion vectors. Then perform alignment and motion compensation at each scale separately to obtain the predictive features on one side. Finally, *concat* and fuse the forward and backward outputs to get the final predictive features F , F_2 and F_3 as prior information for the following residual coding.

Such a fusion strategy not only yields reliable prior information but also allows the subsequent residual coding process to have the exact same model design as the low-delay frameworks, thus simplifying the porting of state-of-the-art P-frame models to the proposed algorithm.

2.3 Scaled Motion Priors from Temporal Contexts

The increase of motion codecs brings more bit rate consumption while improving prediction efficiency. To solve the problem, we utilize the optical flow between forward and backward reference frames prior to further take advantage of bidirectional prediction.

According to the basic assumptions of optical flow method (constant brightness, continuous time or small motion amplitude), it is not difficult to speculate that the optical flow between the frame to be encoded and the reference frame should have a close temporal correlation with the optical flow between the forward and backward reference frames. As shown in Fig. 2, the reference and to-be-coded optical flows in the ParkScene sequence are almost the same for positions with gentle motion such as the background.

As shown in Fig. 1, take forward motion as an example, use the optical flow network to compute the optical flow m_{front} between the frame to be encoded x and the forward reference frame $\hat{x}_{front}$, as well as the optical flow $\tilde{m}_{front}$ between the backward reference frame $\hat{x}_{back}$ and the forward reference frame $\hat{x}_{front}$ during encoding. Take half of the value of $\tilde{m}_{front}$ as prior, and concat it with m_{front} as the input of the motion encoder. When decoding, the output of the motion decoder and $\tilde{m}_{front}$ are jointly input to the motion information reconstruction module to get the final reconstructed motion $\tilde{m}_{front}$.

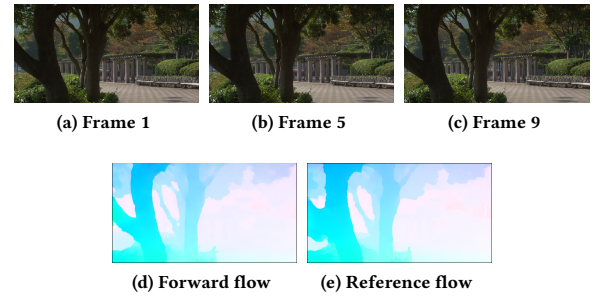


Figure 2: Frames 1, 5, 9 and optical flows of ParkScene in Class B. When encoding Frame 5, Frames 1 and 9 are reference frames. Forward flow refers to flow to be coded between Frame 1 and 5. Reference flow refers to half of optical flow between Frame 1 and 9.

2.4 Trustworthy Motion Modeling

In a hierarchical reference structure, two reference frames are often not in the same layer, which means that the reconstruction quality of them is likely to be quite different. Considering some more practical situations, such as the presence of camera rotation, violent motion and occlusion in the video, may make it difficult for the reference frames on one side to provide valid information. In this case, equal use of the bidirectional reference information is clearly less accurate than greater use of the side with high reconstruction quality.

To this end, the concept of trustworthy motion modeling is introduced to evaluate the effectiveness of reference information. Take forward motion as an example, the output of the motion decoder is expanded and divided into two parts, predictive motion information $\bar{m}_{front}$ and confidence λ_{front}^m . Then adjust the range of values of λ_{front}^m to (0,1) by a sigmoid function.

With this design, the final predicted frame $\bar{x} = \lambda_{front}^m * \bar{x}_{front} + \lambda_{back}^m * \bar{x}_{back}$. Similarly, for the fusion of features, λ_{front}^m is extended by a convolutional neural network and downsampled to obtain $\lambda_{front}^f, \lambda_{front_2}^f$ and $\lambda_{front_3}^f$. Then they are multiplied with the forward prediction features $\bar{F}_{front}, \bar{F}_{front_2}$ and $\bar{F}_{front_3}$, respectively. The same operation is taken for the backward prediction features. Finally, the processed forward and backward reference features are connected to obtain the final prediction features $\bar{F}, \bar{F}_2$ and $\bar{F}_3$ as prior information for residual coding.

2.5 Rate-distortion Optimization Techniques

For hierarchical structures, a reasonable rate allocation strategy within the same GOP plays a crucial role in the overall compression performance. Allocating more code rate to lower level B-frames yields higher reconstruction quality and facilitates more effective reference information for subsequent coded frames. However, using different network structures and parameters for each layer is impractical and introduces excessive additional expenses.

To achieve different compression qualities for different layers of B-frames in a single model, we modify the structure of the codec and the loss function. First, a fully connected function *scalar* is added, whose input is a floating-point number λ_{scalar} between [0,1.4] and the output is a vector. This vector is divided into groups according to the dimensions of the codecs and entropy models to be multiplied with the output of certain layers of the neural network to achieve the code rate adjustment.

In addition, λ_{scalar} is also multiplied with the original λ in the loss function to further adjust the bit rate allocation strategy. For the selection of λ_{scalar} parameter, this scheme refers to the design of RA configuration for bit rate allocation in HEVC to adjust the specific values.

3 Experiments

3.1 Experimental Setup

Datasets. Vimeo-90k[17] and BVI-DVC datasets are used to train our models. Each video of these datasets is randomly cropped into 256×256 patches. We test the coding efficiency using the HEVC Class B-E, UVG [13] and MCL-JCV [15] datasets.

Test conditions. We evaluate our coding performances using the first 96 frames of each sequence. The intra-period and GOP (group of pictures) sizes are set to the same value of 16 or 32. The video quality is evaluated using PSNR and MSSSIM. The BD-rate is used to measure the compression ratio.

Our benchmarks include traditional codecs H.265/HEVC(HM-16.20) and H.266/VTM(VTM-23.0) with random access configuration. Note that the GOP size of HEVC is limited to 16 regardless of the intra-period. B-CANF [2] is picked as the representative scheme for the hierarchical bi-directional prediction models. For uni-directional prediction NVC benchmark, DCVC-HEM, DCVC-DC[8] and DCVC-FM [9] are selected. For YUV420 color space, the BD-Rate calculations are consistent with the standard committee to mix the results of the three color components with weights of 6:1:1(Y:U:V). B-CANF and DCVC-HEM are only optimized for the RGB colorspace, so we convert both origin and reconstructed sequences to YUV420 colorspace when calculating metrics. We also test B-CANF, EEV [5, 6] and our method in RGB colorspace for a more comprehensive comparison.

Model Training. We adopt a multi-stage training strategy. In the pre-train stage, we utilize Vimeo-90K dataset and only train a single model with λ values of 2048 for PSNR. The gradient for each frame is transferred separately. In fine-tuning stage, BVI-DVC is selected because its single sequence contains more frames, which facilitates the model to learn more temporal domain properties. Four models are trained with different λ values of 2048, 512, 256 and 64 for PSNR as well as 64, 32, 16 and 8 for MS-SSIM.

3.2 Comparisons with Previous Methods

Table 1 and table 2 show the BD-rate comparison of PSNR metric for intra-period 16 and 32 in YUV420 colorspace, respectively. The I-frame model we use is migrated from DCVC-FM.

Table 1: BD-Rate(%) comparison for PSNR metric in YUV420 colorspace. 96 frames with intra-period=16. Anchor is HM.

	HEVC Class B-E					UVG	MCL-JCV
	B	C	D	E	Average		
B-CANF	-4.46	9.35	-13.33	0.03	-2.38	13.48	5.49
DCVC-HEM	-8.40	6.40	-5.97	-3.88	-3.24	-2.50	-7.67
DCVC-DC	-24.02	-17.40	-24.85	-24.51	-22.66	-19.41	-35.80
DCVC-FM	-23.41	-20.95	-29.52	-30.38	-25.63	-20.45	-37.76
VTM-23.0	-28.44	-28.74	-23.50	-24.38	-26.52	-32.80	-32.44
Proposed	-20.97	-21.96	-35.39	-25.72	-25.71	5.10	-13.91

Table 2: BD-Rate(%) comparison for PSNR metric in YUV420 colorspace. 96 frames with intra-period=32. Anchor is HM.

	HEVC Class B-E					UVG	MCL-JCV
	B	C	D	E	Average		
B-CANF	9.20	22.52	-6.21	11.20	9.05	31.12	19.29
DCVC-HEM	-6.85	6.80	-7.72	-0.76	-2.51	1.70	-4.28
DCVC-DC	-25.75	-21.01	-30.05	-29.40	-26.32	-21.23	-26.51
DCVC-FM	-29.30	-28.34	-37.65	-38.70	-32.91	-23.20	-28.08
VTM-23.0	-37.81	-35.91	-31.46	-38.02	-35.79	-36.90	-40.10
Proposed	-14.89	-16.60	-34.07	-18.49	-20.79	17.45	-3.41

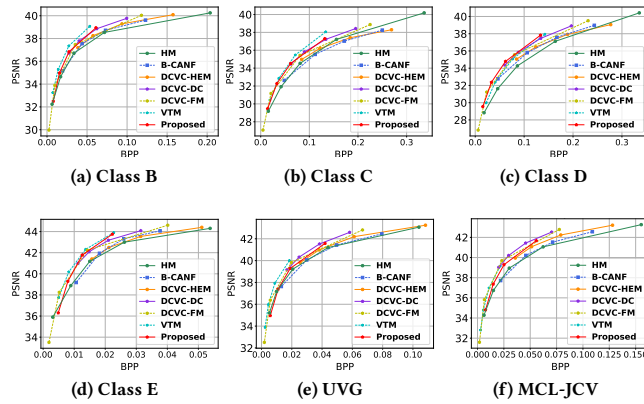


Figure 3: RD curve for PSNR metric in YUV420 colorspace. 96 frames with intra-period=16.

When intra-period is set to be 16, for HEVC test sequences, our scheme achieves comparable performance to state-of-the-art NVC codecs. Regarding UVG and MCL-JCV datasets, the proposed approach has noticeable performance gaps. The main reason lies in longer distance between the low-layer B-frames and their reference frames. For sequences with certain characteristics like violent motion and camera rotation, the reference frames cannot provide effective reference information, making it difficult for the proposed motion capture and feature prediction schemes. For instance, the *Jockey* from UVG with complex movement gets inferior results compared to HEVC.

When the intra-period is set to be 32, the proposed method also far outperforms the existing representative bi-directional prediction model B-CANF on all the test sequences.

Table 3: BD-Rate(%) comparison for PSNR metric in RGB colorspace. 96 frames with intra-period=16. Anchor is the proposed method.

	HEVC Class B-E					UVG	MCL-JCV
	B	C	D	E	Average		
EEV	77.69	70.73	70.21	68.83	72.42	76.31	70.01
B-CANF	12.12	21.23	16.80	33.18	19.52	3.12	13.15

Table 4: BD-Rate(%) comparison for MSSSIM metric in YUV420 colorspace. 96 frames with intra-period=16. Anchor is HM.

	HEVC Class B-E					UVG	MCL-JCV
	B	C	D	E	Average		
B-CANF	195.34	141.20	111.74	62.00	135.90	40.44	18.07
DCVC-HEM	-8.29	-25.00	-34.48	-14.27	-20.14	-55.08	-17.32
DCVC-DC	-8.65	-16.92	-15.57	-18.57	-14.31	-8.43	-21.81
DCVC-FM	-5.66	-15.95	-19.09	-19.70	-14.22	-0.19	-20.94
VTM-23.0	-29.03	-27.79	-24.31	-28.88	-27.51	-31.41	-34.96
Proposed	-16.49	-25.36	-40.13	-19.06	-25.10	10.91	-11.95

To make a fair comparison with B-CANF trained by RGB-loss, tests in RGB colorspace are also carried out. Table 3 shows the BD-rate comparison of PSNR metric for intra-period 16. Without

finetune training for RGB-loss, our model still demonstrates better performance than B-CANF.

Table 4 shows the BD-rate comparison of MS-SSIM metric for intra-period 16 in YUV420 colorspace. The I-frame model we use is migrated from DCVC-HEM. The test results show the same trend as the PSNR metric and our model is still far superior to B-CANF.

4 Conclusion

In conclusion, this paper proposes a scaled hierarchical bi-directional prediction structure to effectively capture temporal correlation among frames to obtain higher coding efficiency. By using parameter-shared motion codecs and efficient information fusion strategies, as well as introducing the concept of trust-worthy motion modeling, our framework can get more accurate prediction features. Scaled motion priors and Interlayer rate-distortion adjustments can further reduce the overall bit rate. The proposed method has been adopted as the latest reference model by MPAAI EEV standard.

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